

A probabilistic model of readiness

Logistic regression as a foundation

Define *readiness* as a probability of successfully completing a task. While this is certainly not a unique definition of readiness, it is a fairly general representation. Assume that this probability depends on normalized scores $s_i \in [0, 1]$, where $i = 1, \dots, n$. Each score reflects some capability or attribute that the task depends on. Like the definition of readiness as a probability, the choice of normalized scores is relatively general.

As a starting point, consider a generalized linear relationship between readiness probability r and the scores s_i :

$$\text{logit}(p) \equiv \ln \frac{r}{1-r} = w_0 + \sum_{i=1}^n w_i s_i , \quad (1)$$

where each weight $w_i \in \mathbb{R}$ measures how much the variation in s_i affects the variation in $\text{logit}(r)$. Here $\text{logit}(r)$ (or *log odds*) is the conventional link function relating a probability to a linear function of independent variables. Note that $\text{logit}(r) \in \mathbb{R}$.

Let $y \in \{0, 1\}$ be a binary variable that denotes the outcome of a single task execution, where $y = 1$ if the task is completed successfully, and 0 otherwise. Suppose there are m task trials, where the outcome y_j and scores s_{ij} are recorded for each trial $j = 1, \dots, m$. Using eq. (1), and assuming that the w_i have normally distributed priors, the corresponding probabilistic model is

$$\begin{aligned} y_j &\sim \text{Bernoulli}(r_j) \\ \text{logit}(r_j) &= w_0 + \sum_{i=1}^n w_i s_{ij} \\ w_i &\sim \text{Normal}(0, \sigma_{w_i}) , \end{aligned} \quad (2)$$

where σ_{w_i} should be appropriately large to reflect high prior uncertainty. If we were to run inference on this model, we would obtain posterior distributions for the w_i . The model in eq. (2) is just logistic regression in probabilistic form. An advantage of writing the model in this way is that it can be easily extended, as shown below.

An error model for scores

Define the scores s_i to be the *true* scores, and the probability r determined by s_i and w_i to be the *true* readiness probability. Of particular interest is the case where the true scores in eq. (2) are not directly observed. Instead, suppose we have observed scores s'_i that have some statistical relationship to the s_i . Since both the observed and true scores are normalized, care must be taken to develop an error model that preserves the normalization. One such error model relates the logits of the two scores:

$$\text{logit}(s'_i) \sim \text{Normal}(\text{logit}(s_i) + b_i, \sigma_{s_i}) , \quad (3)$$

where the term b_i reflects a *bias* for each score. Eq. (3) says that the logit of the observed score is normally distributed around the shifted logit of the true score, with standard deviation σ_{s_i} .

With the addition of the score model, there are new random variables in need of priors. Conventional choices are

$$\begin{aligned}
b_i &\sim \text{Normal}(0, \sigma_{b_i}) \\
\sigma_{s_i} &\sim \text{Exp}(\lambda_i) \\
s_i &\sim \text{Beta}(\alpha_i, \beta_i) ,
\end{aligned} \tag{4}$$

where Beta has support $[0, 1]$, consistent with the normalized score. Both σ_{b_i} and λ_i should be relatively large, while $\alpha_i = \beta_i = 1$ should be used to reflect an initially uniform distribution in the true scores.

Note that the model formulation given here isn't unique. There are other ways to extend the model in depth and breadth. For example, the parameters of the prior distributions in eq. (2) and (4) may be related to each other or to other latent variables.

Decisions under uncertainty

The purpose of a model is to inform decisions. In the case of readiness, we use the model for the probability to decide if a task should be executed based on observed scores $s'_1, \dots, s'_n$. But the probability alone isn't enough to make a decision. For each decision, there are costs associated with possible outcomes. These costs are often subjective and are likely to vary depending on circumstances. This is why probabilities are notoriously difficult to interpret and apply in real-world settings. Despite the ambiguity in weighing outcomes, we can make theoretical headway on the decision-making machinery.

Observed scores have a noisy relationship with the true scores, and weights on the true scores aren't precisely known. Therefore, one set of observed scores corresponds to a distribution of readiness probabilities. Before addressing the full distribution of outcomes, it's instructive to see how a decision can be made based on a single value of the readiness probability.

Let $d \in \{E, N\}$ denote the decision to execute (E) or not (N), and let $o \in \{S, F\}$ denote success (S) or failure (F) of an executed task. For each decision, there is a cost $c(d)$. The cost $c(N)$ of not executing a task might be interpreted as an accrual of long-term consequences of not having the benefits that the task provides. The cost of execution $c(E)$ has two components, one for each outcome:

- $c(S \mid E)$: the cost of successful execution.
- $c(F \mid E)$: the cost of failed execution.

A task that is successfully executed might have significant benefits (negative cost), making it highly desirable to execute the task. However, if the task is executed and results in failure, no benefit is achieved and the failure itself may have a large associated cost. The total cost of execution can be written as the expectation over the outcomes:

$$c(E) = rc(S \mid E) + (1 - r)c(F \mid E) , \tag{5}$$

where r is the readiness probability, and $1 - r$ is the task failure probability.

The decision we should choose is the one that minimizes the cost. Specifically, we should execute the task if $c(E) < c(N)$, which corresponds to a threshold on the readiness probability:

$$r < \frac{c(N) - c(F \mid E)}{c(S \mid E) - c(F \mid E)} . \tag{6}$$

This may look awkward, since we expect there to be a lower bound on r for execution. But note that if $c(E \mid S) < 0$ (benefit), while other cost terms are positive, the decision bound becomes

$$r > \frac{|c(\mathbf{N})| - |c(\mathbf{F} \mid \mathbf{E})|}{|c(\mathbf{S} \mid \mathbf{E})| + |c(\mathbf{F} \mid \mathbf{E})|} . \quad (7)$$

Now let's investigate how these results extend to the general case when we have a distribution of readiness probabilities.

For any probabilistic model, the result of inference is the posterior distribution of model parameters. Let θ represent the collection of all model parameters, and let x represent the collection of all observations. In the model described in previous sections, θ encompasses the weights w_i , biases b_i , etc., while $x = \{(s'_{11}, \dots, s'_{n1}, y_1), \dots\}$, the set of observed score-outcome pairs. The posterior distribution of model parameters given the data can be written formally as $p(\theta \mid x)$. For a random draw from the posterior, the value of $r = r(\theta)$ is known. It follows that the *expected* cost of a decision is¹

$$\mathbb{E}[c(d)] = \int d\theta p(\theta \mid x) c(d \mid \theta) . \quad (8)$$

The appropriate decision is the one that minimizes $\mathbb{E}[c(d)]$.

For this simple problem, it turns out that the form of the decision threshold (see eq. (6)) is unchanged. Note that $\mathbb{E}[c(\mathbf{N})] = c(\mathbf{N})$, and

$$\begin{aligned} \mathbb{E}[c(\mathbf{E})] &= \mathbb{E}[rc(\mathbf{S} \mid \mathbf{E}) + (1 - r)c(\mathbf{F} \mid \mathbf{E})] \\ &= \mathbb{E}[r] c(\mathbf{S} \mid \mathbf{E}) + (1 - \mathbb{E}[r]) c(\mathbf{F} \mid \mathbf{E}) . \end{aligned} \quad (9)$$

It follows that the decision to execute the task corresponds to the condition

$$\mathbb{E}[r] < \frac{c(\mathbf{N}) - c(\mathbf{F} \mid \mathbf{E})}{c(\mathbf{S} \mid \mathbf{E}) - c(\mathbf{F} \mid \mathbf{E})} . \quad (10)$$

Given the posterior $p(\theta \mid x)$, we need only to compute the expected readiness probability in order to make a decision for a certain cost function $c(d)$.

¹This is assuming θ is a continuous space.